

Question 1: Write the following using the appropriate prefix and 3 significant figures.

Vraag 1: Skryf die volgende oor en maak gebruik van die toepaslike voorvoegsel en 3 beduidende syfers. [1]

1.1 62 998 m

63.0 km

Question 2: Determine the following:

Vraag 2: Bepaal die volgende:

[1]

2.1 $\sqrt[3]{5^4}$

1.002

Question 3: Determine the following using your calculator:

Vraag 3: Gebruik jou sakrekenaar om die volgende te bepaal:

[1]

3.1 Log 0.2813 =

-0.551

Question 4:

If the noise level in decibels (dB) is given by the formula $L = 10 \log \frac{I}{I_0}$, where

$I_0 = 10^{-16} \text{ W/cm}^2$, determine the intensity I of sound at $L = 140 \text{ dB}$.

Vraag 4:

Indien die geraasvlak in decibel (dB) gegee word deur die formule $L = 10 \log \frac{I}{I_0}$, waar

$I_0 = 10^{-16} \text{ W/cm}^2$, bepaal die intensiteit I van klank teen $L = 140 \text{ dB}$.

[4]

$$L = 10 \log \frac{I}{I_0}$$

$$14.0 = \log \frac{I}{I_0}$$

$$10^{14} = \frac{I}{10^{-16}}$$

$$I = 10^{-2} \text{ W/cm}^2$$

Question 5: Convert the following to radians.

Vraag 5: Bereken die volgende in radiaal.

[1]

5.1 260°

4.53 rad

Question 6: Convert the following to degrees.

Vraag 6: Bereken die volgende in grade.

[1]

6.1 $2\pi/3 \text{ rad}$

120°

Question 8: A drinking glass has an inner diameter of 3 cm and a height of 15 cm.

Determine the volume of the water which the glass can contain.

Vraag 8: 'n Drinkglas het 'n binne diameter van 3 cm en 'n hoogte van 15 cm. Bereken die volume water wat die glas kan hou. [4]

$$V = Ah = \pi r^2 h = \pi (1.5)^2 \times 15 = 106 \text{ cm}^3$$

Question 9: Determine derivatives of the following function and determine the gradient of the function $x = -1$.

Vraag 9: Bereken die afgeleides van die volgende funksie en bepaal die helling van die funksie by $x = -1$. [2]

$$9.1 \quad y = 3x^2 \sin(5x)$$

$$\frac{dy}{dx} (3x^2 \sin(5x)) = 3x^2 (5) \cos(5x) + \sin(5x) [6x]$$

$$= 15x^2 \cos 5x + 6x \sin 5x$$

$$\frac{dy}{dx} (3x^2 \sin(5x)) = 15x^2 \cos 5x + 6x \sin 5x$$

When $x = -1$:

$$\frac{dy}{dx} = 15 \cos(-5) + 6 \sin(-5) = 14.9 + 0.52$$

$$= 15.42$$

Question 10: What is the sum of the following four vectors in unit vector notation?

Vraag 10: Bepaal die som van die volgende vier vektore.

[10]

$\vec{A}$: 6.00 m at +0.90 rad

$\vec{C}$: 5.00 m at -75°

$\vec{B}$: 4.00 m at +1.20 rad

$\vec{D}$: 6.00 m at -210°

$$A_x = 3.73 \text{ m} \quad A_y = 4.70 \text{ m}$$

$$B_x = 1.45 \text{ m} \quad B_y = 3.73 \text{ m}$$

$$C_x = 1.29 \text{ m} \quad C_y = -4.83 \text{ m}$$

$$D_x = -5.20 \text{ m} \quad D_y = 3.00 \text{ m}$$

$$R_x = 1.27 \text{ m} \quad R_y = 6.60 \text{ m}$$

$$\vec{R} = 1.27\hat{i} + 6.60\hat{j}$$

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Question 1: Write the following using the appropriate prefix and 3 significant figures.

Vraag 1: Skryf die volgende oor en maak gebruik van die toepaslike voorvoegsel en 3 beduidende syfers. [1]

1.1 166 998 m

167 km

Question 2: Determine the following:

Vraag 2: Bepaal die volgende: [1]

2.1 $\sqrt[3]{7^{0.2}}$

1.139

Question 3: Determine the following using your calculator:

Vraag 3: Gebruik jou sakrekenaar om die volgende te bepaal: [1]

3.1 Log 0.789 =

-0.103

Question 4:

If the noise level in decibels (dB) is given by the formula $L = 10 \log \frac{I}{I_0}$, where

$I_0 = 10^{-16} \text{ W/cm}^2$, determine the intensity I of sound at $L = 105 \text{ dB}$.

Vraag 4:

Indien die geraasvlak in decibel (dB) gegee word deur die formule $L = 10 \log \frac{I}{I_0}$, waar

$I_0 = 10^{-16} \text{ W/cm}^2$, bepaal die intensiteit I van klank teen $L = 105 \text{ dB}$. [4]

$$L = 10 \log \frac{I}{I_0}$$

$$10.5 = \log \frac{I}{I_0}$$

$$10^{10.5} = \frac{I}{10^{-16}}$$

$$I = 10^{-5.5} \text{ W/cm}^2$$

Question 5: Convert the following to radians.

Vraag 5: Bereken die volgende in radiaal. [1]

5.1 245°

$$4.28 \text{ rad} = \frac{49}{36} \pi$$

Question 6: Convert the following to degrees.

Vraag 6: Bereken die volgende in grade. [1]

6.1 $5\pi/3 \text{ rad}$

300°

Question 8: A drinking glass has an inner diameter of 3 cm, an outer diameter of 3.5 cm and a height of 15 cm. Determine the volume of the glass which is needed to make the drinking glass.

Vraag 8: 'n Drinkglas het 'n binne diameter van 3 cm, 'n buite diameter van 3.5 cm en 'n hoogte van 15 cm. Bereken die volume glas wat nodig is om die drinkglas te produseer. [4]

Volume of glass = inner volume – outer volume

$$\begin{aligned} V &= A_o h_o - A_i h_i = \pi r_o^2 h_o - \pi r_i^2 h_i \\ &= \left(\pi (1.75)^2 \times 15 \right) - \left(\pi (1.5)^2 \times 14.75 \right) \\ &= 40.06 \text{ cm}^3 \end{aligned}$$

Note that the heights must differ by the thickness of the glass. If the $h_i = 15 \text{ cm}$ then h_o must be 15.25 cm then $V = 40.69 \text{ cm}^3$

Question 9: Determine derivatives of the following function and determine the gradient of the function $x = -1$.

Vraag 9: Bereken die afgeleides van die volgende funksie en bepaal die helling van die funksie by $x = -1$. [2]

$$9.1 \quad y = (3x^2 + 5x)(2x^3 - 3)$$

$$\begin{aligned} y &= (3x^2 + 5x)(2x^3 - 3) \\ y &= 6x^5 - 9x^2 + 10x^4 - 15x \end{aligned}$$

$$\frac{dy}{dx} = 30x^4 + 40x^3 - 18x - 15$$

$$\text{When } x = -1: \quad \frac{dy}{dx} = -7$$

Question 10: Determine the sum of the following three vectors in unit vector notation and also determine the magnitude and direction of the vector.

Vraag 10: Bepaal die som van die volgende drie vektore in eenheids-vektor notasie, en bepaal ook die grootte en rigting van die vektor. [10]

$\vec{A}$: 6.00 m at $+0.90 \text{ rad}$

$\vec{C}$: 5.00 m at -75°

$\vec{D}$: 6.00 m at -210°

$$A_x = 3.73 \text{ m} \quad A_y = 4.70 \text{ m}$$

$$C_x = 1.29 \text{ m} \quad C_y = -4.83 \text{ m}$$

$$D_x = -5.20 \text{ m} \quad D_y = 3.00 \text{ m}$$

$$R_x = -0.18 \text{ m} \quad R_y = 2.87 \text{ m}$$

$$\vec{R} = -0.18\hat{i} + 2.87\hat{j}$$

$$|\vec{R}| = \sqrt{R_x^2 + R_y^2} = \sqrt{(-0.18)^2 + 2.87^2} = 2.88 \text{ m}$$

$$\text{Direction: } \theta = \tan^{-1} \left(\frac{0.18}{2.87} \right) = 86.4^\circ \text{ above negative } x\text{-axis (} 93.6^\circ \text{ anticlockwise from } +x\text{-axis)}$$

Continue your answer over the page

Question 1: Write the following using the appropriate prefix and 3 significant figures.

Vraag 1: Skryf die volgende oor en maak gebruik van die toepaslike voorvoegsel en 3 beduidende syfers. [1]

1.2 1 998 m

2.00 m

Question 2: Determine the following:

Vraag 2: Bepaal die volgende: [1]

2.1 $\frac{3^4}{4^3}$

1.27

Question 3: Determine the following using your calculator:

Vraag 3: Gebruik jou sakrekenaar om die volgende te bepaal: [1]

3.1 $e^{5.14} =$

170.7

Question 4:

If the noise level in decibels (dB) is given by the formula $L = 10 \log \frac{I}{I_0}$, where

$I_0 = 10^{-16} \text{ W/cm}^2$, determine the noise level if the intensity of the sound at a point is measured as $I_0 = 10^{25} \text{ W/cm}^2$.

Vraag 4:

Indien die geraasvlak in decibel (dB) gegee word deur die formule $L = 10 \log \frac{I}{I_0}$, waar

$I_0 = 10^{-16} \text{ W/cm}^2$, bepaal die geraasvlak indien die intensiteit $I = 10^{25} \text{ W/cm}^2$ word gemeet by 'n sekere punt. [4]

$$\begin{aligned} L &= 10 \log \frac{I}{I_0} \\ &= 10 \log \frac{10^{25}}{10^{-16}} \\ &= 10 \log 10^{41} \\ L &= 410 \text{ dB} \end{aligned}$$

Question 5: Convert the following to radians.

Vraag 5: Bereken die volgende in radiaal. [1]

5.2 12°

0.209 rad

Question 6: Convert the following to degrees.

Vraag 6: Bereken die volgende in grade. [1]

6.2 0.6 rad

34.4°

Question 8: A drinking glass has an inner diameter of 3 cm, an outer diameter of 3.5 cm and a height of 15 cm. Determine the volume of the glass which is needed to make the drinking glass.

Vraag 8: 'n Drinkglas het 'n binne diameter van 3 cm, 'n buite diameter van 3.5 cm en 'n hoogte van 15 cm. Bereken die volume glas wat nodig is om die drinkglas te produseer. [4]

Volume of glass = inner volume – outer volume

$$\begin{aligned} V &= A_o h_o - A_i h_i = \pi r_o^2 h_o - \pi r_i^2 h_i \\ &= \left(\pi (1.75)^2 \times 15 \right) - \left(\pi (1.5)^2 \times 14.75 \right) \\ &= 40.06 \text{ cm}^3 \end{aligned}$$

Note that the heights must differ by the thickness of the glass. If the $h_i = 15 \text{ cm}$ then h_o must be 15.25 cm and the Volume will be slightly more than above.

Question 9: Determine derivatives of the following function and determine the gradient of the function $x = -1$.

Vraag 9: Bereken die afgeleides van die volgende funksie en bepaal die helling van die funksie by $x = -1$. [2]

$$9.1 \quad y = (3x^2 + 5x)(\cos 2x)$$

$$\begin{aligned} y &= (3x^2 + 5x)(\cos 2x) \\ \frac{dy}{dx} &= (3x^2 + 5x)(-2 \sin 2x) + (\cos 2x)(6x + 5) \\ &= (-6x^2 - 10x)(\sin 2x) + (6x + 5)(\cos 2x) \end{aligned}$$

$$\text{When } x = -1: \quad \frac{dy}{dx} = (4)(\sin(-2)) + (-1)(\cos(-2))$$

$$\frac{dy}{dx} = 4(-0.0348) + 1(-0.999) = -1.13$$

Question 10: Determine the sum of the following three vectors in unit vector notation and also determine the magnitude and direction of the vector.

Vraag 10: Bepaal die som van die volgende drie vektore in eenheids-vektor notasie, en bepaal ook die grootte en rigting van die vektor. [10]

$$\vec{A}: 4.00 \text{ m at } +0.90 \text{ rad} \quad \vec{C}: 5.00 \text{ m at } -75^\circ$$

$$\vec{B}: 6.00 \text{ m at } +1.20 \text{ rad}$$

$$A_x = 2.49 \text{ m} \quad A_y = 3.13 \text{ m}$$

$$B_x = 2.17 \text{ m} \quad B_y = 5.59 \text{ m}$$

$$C_x = 1.29 \text{ m} \quad C_y = -4.83 \text{ m}$$

$$R_x = 5.95 \text{ m} \quad R_y = 3.89 \text{ m} \quad \vec{R} = 5.95\hat{i} + 3.89\hat{j}$$

$$|\vec{R}| = \sqrt{R_x^2 + R_y^2} = \sqrt{5.95^2 + 3.89^2} = 7.11 \text{ m}$$

$$\text{Direction: } \theta = \tan^{-1} \left(\frac{3.89}{5.95} \right) = 33.1^\circ \text{ above positive } x\text{-axis}$$